

Danielle Tsao

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EDUCATION

University of Washington

PhD, Statistics

Seattle, WA, USA

2023-present

University of Victoria

B.Sc. Honours, Statistics (GPA 8.98/9.0)

Victoria, BC, Canada

2019-2023

- Select courses: [Mathematics and Statistics] Real Analysis, Probability Theory, Mathematical Statistics, Multivariate Analysis, Machine Learning, Stochastic Processes, Linear Algebra, Statistical Computing
- Select courses: [Computer Science] Database Design and Implementation, Software Development Methods, Algorithms and Data Structures

PUBLICATIONS

- **Danielle Tsao**, Ronan Perry, and Carlos Cinelli (2025). "On the minimum strength of (unobserved) covariates to overturn an insignificant result." *arXiv:2408.13901*. To appear in *Statistical Science*.
- Xiangling Ji, **Danielle Tsao**, Kailun Bai, Min Tsao, Li Xing, and Xuekui Zhang (2023). "scAnnotate: an automated cell-type annotation tool for single-cell RNA-sequencing data." *Bioinformatics Advances*, 3(1), p.vbad030. DOI: 10.1093/bioadv/vbad030

RESEARCH EXPERIENCE

Research Assistant, Statistics, University of Washington

2025-2026

Supervisor: Emilija Perković (University of Washington)

- Studied the causal interpretation of the instrumental variable estimand under an aggregate treatment.

Research Assistant, Statistics, University of Washington

2024-2025

Supervisor: Carlos Cinelli (University of Washington)

- Studied reversals of insignificance and robust statistical inference of linear regression coefficients.

Research Assistant, Biostatistics, Mitacs Accelerate Internship Program

2022-2023

Supervisor: Xuekui Zhang (University of Victoria)

- Designed an updated guideline for cellular deconvolution on RNA-seq.

Research Assistant, Biostatistics, University of Victoria

2022

Supervisor: Xuekui Zhang

- Led a summer research team through the development of a deconvolution benchmarking pipeline in R.

TEACHING

Teaching Assistant, Statistics, University of Washington

2023-2025

- STAT 220: Statistical Reasoning (Fall 2023, Summer 2024), STAT 395: Probability II (Winter 2024), STAT 311: Elements of Statistical Methods (Summer 2025), STAT 502: Design and Analysis of Experiments (Winter 2025)

Teaching Assistant, Mathematics and Statistics, University of Victoria

2022

- Tutored and graded undergraduate math and statistics courses in the Math and Stats Assistance Centre.

ACADEMIC SERVICE

- Workflow Chair** for the 42nd Conference on Uncertainty in Artificial Intelligence 2026
- Member** of the Statistics Outreach and Community Committee, University of Washington 2024 - present
- Member** of the Association for Women in Math and Statistics, University of Victoria 2020-2023
- Participated in outreach to women in math and statistics by speaking at annual new student orientations.

PRESENTATIONS AND RESEARCH STAYS

- **Causal inference: From theory to practice and back again (CIF)**, Cambridge, UK. Hosted by the Isaac Newton Institute of Mathematical Sciences. Invited Participant.
- **CIF Workshop on Foundations of Causal Inference**, Cambridge, UK, “Lost in Aggregation: The Causal Interpretation of the IV Estimand”, joint work with Emilija Perković (advisor), Krikamol Muandet and Frederick Eberhardt, Poster Presentation.
- **Jamie Cassels Undergraduate Research Award Poster Fair 2023**, Victoria, Canada. "scAnnotate: an automated cell-type annotation tool for single-cell RNA-sequencing data", joint work with Xiangling Ji, Kailun Bai, Min Tsao, Li Xing, and Xuekui Zhang, Poster Presentation.
- **Canadian Statistics Student Conference (CSSC) 2022**, "scAnnotate: an automated cell-type annotation tool for single-cell RNA-sequencing data", joint work with Xiangling Ji, Kailun Bai, Min Tsao, Li Xing, and Xuekui Zhang, Poster Presentation.

HONOURS AND AWARDS

American Statistical Association

Gertrude M. Cox Scholarship Honourable Mention 2025

University of Victoria

- University of Victoria Faculty of Science Jubilee Medal 2023
- University of Victoria Excellence Scholarship (\$26,000) 2019-2023
- Faculty of Science Dean's List 2019-2023
- Science Emerging Researcher Award (\$5,500) 2022
- Jamie Cassels Undergraduate Research Award in Mathematics and Statistics (\$1,500) 2022
- Mathematics and Statistics Scholarship (\$500) 2022
- NSERC Undergraduate Student Research Award in Statistics (\$6,000) 2021
- Mathematics and Statistics Angus & Annie MacKay Scholarship (\$1,700) 2021
- Mathematics and Statistics Donald & Evelyn Munro Scholarship (\$1,000) 2019
- Excellence in Math Scholarship (\$2,500) 2019

Canadian Statistics Student Conference

- Best Undergraduate Poster Presentation 2022

Canadian Mathematical Society

- CMS-RBC Excellence in Mathematics Scholarship (\$1,000) 2021

Performing Arts

- Featured soloist with:
 - Victoria Symphony – Symphony Splash 2018
 - Sooke Philharmonic Orchestra – Young Masters Concert Series 2018